Recovering Sparse Neural Connectivity from Partial Measurements: A Covariance-Based Approach with Granger-Causality Refinement

Quilee Simeon
Massachusetts Institute of Technology
qsimeon@mit.edu

Abstract

Inferring the connectivity of neural circuits from incomplete observations is a fundamental challenge in neuroscience. We present a covariance-based method for recovering the weight matrix of a recurrent neural network from partial measurements across multiple recording sessions, accumulating pairwise covariance estimates from sessions with overlapping neuron subsets. A Granger-causality refinement step enforces biological constraints via projected gradient descent, achieving strong correlation with true connectivity and substantial improvement over chance baselines using partial neuron coverage per session. Through systematic experiments on synthetic networks modeling neural circuits, we characterize a fundamental *control-estimation tradeoff*: stimulation aids identifiability but disrupts intrinsic dynamics, with the optimal level depending on measurement density. We discover that the "incorrect" linear approximation acts as implicit regularization—matching or outperforming the oracle estimator with known nonlinearity at all tested regimes—a concrete instance of James–Stein shrinkage. Error decomposition reveals that correlation from central pattern generators, not model mismatch, is the dominant error bottleneck, identifying intrinsic dynamics modeling as the primary target for further improvement.

1 Introduction

A central goal in systems neuroscience is to map the connectivity of neural circuits from functional measurements [Power et al., 2011]. In neural circuits—from the lobster stomatogastric ganglion ($\sim$30 neurons; [Marder and Bucher, 2007]), to *C. elegans* (302 neurons; [White et al., 1986, Cook et al., 2019]), larval zebrafish [Ahrens et al., 2013], and cortical organoids [Quadrato et al., 2017]—the number of neurons ranges from tens to 10^4 but simultaneous recording of the entire circuit remains difficult. This creates a fundamental inverse problem: given partial, noisy measurements of neural dynamics across multiple sessions, can we recover the full connectivity matrix?

This problem embodies a *control-estimation tradeoff* related to classical notions of persistent excitation in system identification [Ljung, 1998]: stimulating neurons with random perturbations aids connectivity estimation by exciting network modes, but disrupts the intrinsic dynamics that make the circuit biologically relevant. An ideal experimental protocol must balance identifiability against preservation of natural dynamics. Furthermore, real circuits possess intrinsic oscillatory patterns—central pattern generators (CPGs) [Marder and Bucher, 2001]—whose detailed structure is unknown to the experimenter. These represent *unmodeled autonomous dynamics* that influence the observed activity.

Preprint. Under review.

We formalize this as a matrix recovery problem from partial observations. Consider a network of N neurons with connectivity matrix $W \in \mathbb{R}^{N\times N}$, governed by discrete-time recurrent dynamics [Sussillo and Abbott, 2009], where W_{ij} represents the synaptic weight from neuron j to neuron i. In each recording session, we observe a subset of neurons and stimulate a (possibly different) subset. Our key insight is that pairwise covariance statistics can be accumulated across sessions: whenever neurons i and j are simultaneously observed, their covariance contributes to estimating the (i, j) block of W.

Contributions. Our method is general, applying to any network where the same connectivity is measured across multiple partial-observation sessions.

- We derive a covariance-based estimator $\hat{W} = \Sigma_{x_{t+1},x_t}\Sigma^{-1}_{x_t,x_t}$ for connectivity recovery from partial measurements and provide identifiability conditions.
- We introduce a Granger-causality-inspired [Granger, 1969] refinement via projected gradient descent that enforces Granger non-causality sparsity, complementing structural priors (no autapses, non-negativity) applied uniformly to all estimates.
- We discover an *implicit regularization* effect: the "incorrect" linear approximation yields a better-conditioned estimator than the oracle with known nonlinearity, analogous to James–Stein shrinkage [James and Stein, 1961].
- We systematically characterize the tradeoff between stimulation strength and measurement density, revealing that optimal stimulation depends on the fraction of observed neurons.

2 Problem Formulation

2.1 Dynamical System Model

We model the neural circuit as a discrete-time recurrent dynamical system:

$$x_{t+1} = W\phi(x_t) + b_t \tag{1}$$

where $x_t \in \mathbb{R}^N$ is the neural state, $W \in \mathbb{R}^{N\times N}$ is the connectivity matrix, $\phi : \mathbb{R} \to \mathbb{R}$ is an element-wise nonlinearity, and $b_t \in \mathbb{R}^N$ is the total input comprising both extrinsic stimulation and intrinsic central pattern generator (CPG) drive. This arises from discretizing a continuous-time neural circuit $\tau\, d\mathbf{x}/dt = -\tilde{M}\phi(\mathbf{x}) + \tilde{\mathbf{b}}$ with time constant τ and coupling matrix $\tilde{M}$, yielding $W = I - (\Delta t/\tau)\tilde{M}$.

Modeling priors on network topology and weights. We impose three structural priors on W, each a modeling choice motivated by practical and mathematical considerations:

(i) No autapses: $W_{ii} = 0$ for all i. We hypothesize that self-connections are not a necessary feature of the circuits we model, for two reasons. First, autapses appear to be rare in connectomics reconstructions: to our knowledge, most electron microscopy (EM) studies across organisms (*C. elegans*, *Drosophila*, mouse cortex) do not report self-synapses as a distinct structural feature. Second, any self-loop on neuron i can be equivalently modeled by a reciprocating pair $i \to j \to i$ for some neighbor j, making explicit autapses redundant in a strongly connected graph. In our estimation pipeline, this prior is applied as a *known structural constraint*: we zero the diagonal of $\hat{W}$ before any refinement. This is a qualitatively different use of prior knowledge than the "oracle" that knows the nonlinearity ϕ (Section 4); it is an experimenter's belief about circuit architecture, not a biophysical measurement.

(ii) Strong connectivity: We require the digraph to be strongly connected, ensuring that for any node i and any stimulated set $\mathcal{S}$, there exists a directed path from some $s \in \mathcal{S}$ to i. This is a practical requirement for identifiability: in a weakly connected graph, sink components receive no stimulation signal, leaving their covariance rows rank-deficient (see Appendix A.4). Empirically, weakly connected topologies produced dead zones where activity collapsed to zero, while strong connectivity ensured signal reverberation through the entire network.

(iii) Non-negative weights: $W_{ij} \geq 0$. Like the no-autapse prior, this is a modeling belief applied uniformly to *all* estimates—both the raw covariance estimator and the Granger-refined output—not a constraint specific to any refinement step. It is motivated by the measurement modality: EM connectomics, the primary source of ground-truth wiring data, measures synapse *size* (contact area, vesicle

count) but not synapse *sign* (excitatory vs. inhibitory), yielding an inherently unsigned weight matrix. In our formulation, inhibitory effects arise from the *sign of the activity*: a positive weight applied to negative (hyperpolarized) presynaptic activity drives the postsynaptic neuron downward. This is a dual of the standard neuroscience convention (non-negative firing rates with signed weights), and is functionally equivalent for the covariance estimator, which assumes no sign structure on W. By the Perron–Frobenius theorem, non-negative irreducible matrices have a unique maximal real eigenvalue with a positive eigenvector, which simplifies spectral radius control and ensures stable dynamics. The method generalizes to mixed-sign weights (see Section 6).

Nonlinearity. We require the element-wise nonlinearity ϕ to be *1-Lipschitz*: $|\phi'(x)| \leq 1$ for all x. Combined with $\rho(W) \leq 1$, this guarantees contractivity of the autonomous dynamics. Our default is $\phi = \tanh$, which is 1-Lipschitz, bounded, and odd-symmetric ($\tanh(-x) = -\tanh(x)$, so $\tanh(0) = 0$); these additional properties enable exact error analysis via the Stein–Price identity (Section 3). To assess robustness, we test three additional nonlinearities in E5 (Section 4): *identity* ($\phi(x) = x$, 1-Lipschitz but unbounded), *ReLU* ($\phi(x) = \max(0, x)$, 1-Lipschitz but neither odd nor bounded), and *sigmoid* ($\phi(x) = 1/(1 + e^{-x})$, 1-Lipschitz and bounded but not odd, with $\phi(0) = 0.5 \neq 0$). This deliberately includes nonlinearities that violate our default's symmetry and boundedness, testing how gracefully the estimator degrades.

Stability. For stability, we require $\rho(W) \leq 1$, where $\rho(\cdot)$ denotes the spectral radius. Combined with the 1-Lipschitz property of ϕ, this ensures bounded autonomous dynamics ($b_t = 0$): since $\|\phi(x)\| \leq \|x\|$ and $\rho(W) \leq 1$, the map $x \mapsto W\phi(x)$ is non-expansive. The spectral radius is scaled to $\rho(W) \approx 1.0$ (edge of stability), producing marginally stable dynamics that are neither trivially decaying nor divergent. Crucially, with $\rho(W) \leq 1$ and 1-Lipschitz ϕ satisfying $|\phi(x)| < |x|$ for $x \neq 0$ (as with $\tanh$), the autonomous system ($b_t = 0$) *contracts to the origin*: the network falls silent without input. This motivates the central pattern generator (CPG) component of the input b_t: a subset of neurons possess autonomous oscillatory drive (analogous to cardiac pacemaker cells) that sustains network activity in the absence of external stimulation [Marder and Bucher, 2001]. The input thus decomposes as $b_t = b_t^{\text{stim}} + b_t^{\text{CPG}}$, where b_t^{stim} is extrinsic i.i.d. Gaussian noise (independent of x_t) and $b_t^{\text{CPG}} = \text{cpg}(x_t)$ is state-dependent intrinsic drive with $\text{Cov}(b_t^{\text{CPG}}, x_t) \neq 0$. This decomposition is critical: the extrinsic component satisfies our independence assumption, while the CPG component violates it and dominates the estimation error (Section 6).

Crucially, CPG identity is hidden from the estimation pipeline. Which neurons possess intrinsic drive is a property of the ground-truth simulation, not available to the observer. In each session, any measured or stimulated node may also be a CPG node; the experimenter does not know. The variance-based CPG detection described in Section 6 is a *post-hoc* analysis, not an oracle input to the estimator. This mirrors the biological reality: a real experimenter typically cannot identify pacemaker neurons *a priori*.

2.2 Observation Model

In session k, we observe a subset $\mathcal{M}_k \subseteq \{1, \ldots, N\}$ of neurons and stimulate a subset $\mathcal{S}_k$. The observed activity is:

$$y_t^{(k)} = P_{\mathcal{M}_k} x_t + \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, \sigma_\epsilon^2 I) \tag{2}$$

where $P_{\mathcal{M}_k}$ is a projection onto the observed neurons and ϵ_t is i.i.d. measurement noise. Our experiments test both noiseless ($\sigma_\epsilon = 0$) and noisy ($\sigma_\epsilon > 0$) regimes; the method is robust to moderate noise (Section 6).

2.3 The Estimation Problem

Given K sessions with observations $\{y_t^{(k)}\}_{t=1}^{T_k}$ for $k = 1, \ldots, K$, recover W.

3 Methods

3.1 Covariance-Based Estimator

Under the linear approximation $\phi(x) \approx x$ (valid when $\|x\|$ is small), Eq. (1) simplifies to $x_{t+1} \approx Wx_t + b_t$. Assuming statistical independence between inputs b_t and states x_t (an approximation we

revisit critically in Section 6), the cross-covariance of the *true* states satisfies:

$$\Sigma_{x_{t+1},x_t} = W\Sigma_{x_t,x_t} + \underbrace{\Sigma_{b_t,x_t}}_{\approx 0} \tag{3}$$

In the noiseless case ($\sigma_\epsilon = 0$, so $y_t = x_t$), this directly yields the estimator $\hat{W} = \Sigma_{y_{t+1},y_t}\Sigma^{-1}_{y_t,y_t}$. When observation noise is present ($\sigma_\epsilon > 0$), the observed covariances differ from the true state covariances. Since ϵ_t is i.i.d. and independent of x_t:

$$\Sigma_{y_t,y_t} = \Sigma_{x_t,x_t} + \sigma_\epsilon^2 I \tag{4}$$

$$\Sigma_{y_{t+1},y_t} = \Sigma_{x_{t+1},x_t} \quad \text{(noise at } t{+}1 \text{ independent of } y_t\text{)} \tag{5}$$

The estimator computed from observed data is therefore:

$$\hat{W} = \Sigma_{y_{t+1},y_t}\big(\Sigma_{y_t,y_t}\big)^{-1} = W\Sigma_{x,x}\big(\Sigma_{x,x} + \sigma_\epsilon^2 I\big)^{-1} \tag{6}$$

This is a *ridge-regularized* version of the true W: observation noise shrinks the estimate toward zero, with σ_ϵ^2 playing the role of the ridge parameter. For small noise ($\sigma_\epsilon \ll \sigma_{\min}(\Sigma_{x,x})$), the effect is negligible; for large noise, the estimate is over-regularized.

No-autapse prior. Before any further refinement, we enforce the structural prior $\hat{W}_{ii} = 0$ for all i. This is a case of the experimenter applying a *modeling belief* about the underlying circuit (namely, that autapses are absent), and this belief happens to be correct because we constructed the ground truth this way. The diagonal of the raw estimate $\hat{\Sigma}_{y_{t+1},y_t}\hat{\Sigma}^{-1}_{y_t,y_t}$ is dominated by single-neuron autocorrelation (3–4$\times$ larger than off-diagonal entries), which does not reflect genuine self-connectivity. Zeroing the diagonal removes this artifact and is the single most impactful step in the pipeline: in ablation experiments (E2, Section 4), it accounts for the majority of the improvement over the chance baseline. Note that this prior is qualitatively different from the oracle knowledge tested in E6: it is a structural assumption about circuit architecture, whereas E6 tests knowledge of the neuronal transfer function ϕ, a biophysical property that would require single-cell measurements to determine.

Accumulation across sessions. For each pair (i, j), we compute empirical covariances from the *observed* data y_t only in sessions where both neurons i and j are simultaneously measured. Let $S_{ij}^{(k)} = \mathbf{1}[i \in \mathcal{M}_k \wedge j \in \mathcal{M}_k]$ indicate co-measurement. The accumulated covariance is:

$$\hat{\Sigma}_{ij} = \frac{\sum_{k=1}^{K} S_{ij}^{(k)}\hat{\Sigma}_{ij}^{(k)}}{\sum_{k=1}^{K} S_{ij}^{(k)}} \tag{7}$$

where $\hat{\Sigma}_{ij}^{(k)}$ is the empirical covariance of the observed y_t from session k.

Error analysis. The estimation error decomposes into two terms:

$$\hat{W} - W = \underbrace{W\big[\Sigma_{\phi(x),x} - \Sigma_{x,x}\big]\Sigma^{-1}_{x,x}}_{E_1\text{: model mismatch}} + \underbrace{\Sigma_{b,x}\Sigma^{-1}_{x,x}}_{E_2\text{: input correlation}} \tag{8}$$

For $\phi = \tanh$ and Gaussian states $x \sim \mathcal{N}(0, \Sigma)$, the Stein–Price identity [Price, 1958, Stein, 1981] gives an *exact* characterization. Since $\mathbb{E}[\tanh(x_i)x_j] = \Sigma_{ij}\,\mathbb{E}[\mathrm{sech}^2(x_i)]$, we have:

$$\Sigma_{\phi(x),x} = D\Sigma_{x,x}, \qquad D = \mathrm{diag}\big(\mathbb{E}[\mathrm{sech}^2(x_i)]\big) \tag{9}$$

where $0 < D_{ii} < 1$ for $\sigma_i > 0$. Since $E_1 = W(D - I)\Sigma_{x,x}\Sigma^{-1}_{x,x} = W(D - I)$, the mismatch error simplifies to $\|E_1\|_F = \|W(D - I)\|_F \le \|W\|_F \max_i(1 - \mathbb{E}[\mathrm{sech}^2(x_i)])$. For small state variance, $1 - \mathbb{E}[\mathrm{sech}^2(x_i)] \approx \sigma_i^2$, so the error is $O(\max_i \sigma_i^2)$, substantially tighter than generic Lipschitz bounds. Crucially, $\|\Sigma^{-1}_{x,x}\|_2$ amplifies both error sources: ill-conditioned covariance (from sparse measurement) degrades recovery even with perfect model knowledge.

3.2 Granger-Causality Refinement

The raw estimator $\hat{W}$ already enforces the structural priors from Section 2 (no autapses, non-negativity). The Granger refinement adds one additional constraint via projected gradient descent:

1. **No self-connections**: $W_{ii} = 0$ for all i (enforced on all estimates; re-enforced here for consistency).
2. **Granger non-causality**: $W_{ij} = 0$ where $\Sigma_{x_t,x_t}(i,j) > \Sigma_{x_{t+1},x_t}(i,j)$, indicating that the lagged covariance does not exceed the contemporaneous covariance. This sparsity pattern is the unique contribution of the Granger step.

Non-negativity ($W_{ij} \geq 0$) is a modeling prior applied to all estimates (Section 2), not specific to the Granger refinement.

The optimization solves:

$$\min_A \|A - \hat{\Sigma}_{x_{t+1},x_t}\|_F^2 \quad \text{s.t.} \quad A\hat{\Sigma}_{x_t,x_t}^{-1} \in \mathcal{C} \tag{10}$$

where $\mathcal{C}$ is the constraint set. We alternate between gradient steps on A and projection of $W = A\hat{\Sigma}_{x_t,x_t}^{-1}$ onto $\mathcal{C}$.

3.3 Full Pipeline

The complete method is summarized in Algorithm 1 and illustrated in Figure 1.

Algorithm 1 Covariance Accumulation + Granger Refinement

Require: Sessions $\{(y_t^{(k)}, \mathcal{M}_k)\}_{k=1}^K$
Ensure: Estimated connectivity $\hat{W}$

```
Initialize Σ̂_{y_{t+1},y_t} ← 0, Σ̂_{y_t,y_t} ← 0, C ← 0 {accumulators}
for k = 1, ..., K do
    Compute session covariances from observed data {y_t^(k)}_{t=1}^{T_k}
    S ← 1_{M_k} 1_{M_k}^T {co-measurement mask}
    Σ̂_{y_{t+1},y_t} += S ⊙ Σ̂^(k)_{y_{t+1},y_t};   Σ̂_{y_t,y_t} += S ⊙ Σ̂^(k)_{y_t,y_t}
    C += S
end for
Σ̂_{y_{t+1},y_t} ← Σ̂_{y_{t+1},y_t} ⊘ C;   Σ̂_{y_t,y_t} ← Σ̂_{y_t,y_t} ⊘ C {average}
Ŵ_approx ← Σ̂_{y_{t+1},y_t} · Σ̂†_{y_t,y_t} {ridge-regularized estimator, Eq. (6)}
diag(Ŵ_approx) ← 0;   Ŵ_approx ← max(Ŵ_approx, 0) {structural priors: no autapses, non-negativity}
Ŵ ← GrangerRefine(Ŵ_approx, Σ̂_{y_t,y_t}, Σ̂_{y_{t+1},y_t}, C) {Granger sparsity, Eq. (10)}
return Ŵ
```

4 Experiments

4.1 Experimental Setup

We generate synthetic networks modeling neural circuits at biologically motivated scales:

- Random directed graphs with non-negative weights and spectral radius $\rho(W) \leq 1$
- Intrinsic dynamics driven by central pattern generators (CPGs) modeled as chaotic reservoir networks
- Extrinsic stimulation as *independent* i.i.d. Gaussian noise applied to each stimulated node at each timestep (the most favorable case for identifiability; correlated or shared stimulation would reduce the rank of the input covariance)

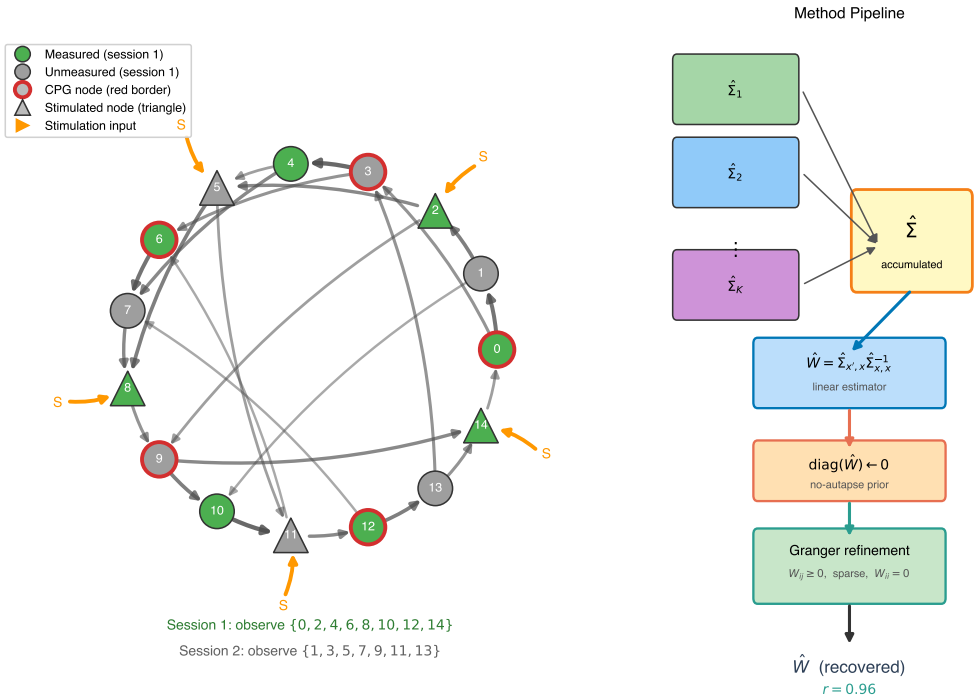

Figure 1: **Problem setup and estimation pipeline. (Left)** A network of $N = 15$ neurons with directed weighted connectivity W. In a given session, a subset of neurons (green) is measured while the rest (gray) are unobserved. Red-bordered nodes are CPG drivers (intrinsic oscillatory input); triangle nodes receive extrinsic stimulation (orange arrows "S"). **(Right)** The method pipeline: each of K sessions yields a partial covariance matrix $\hat{\Sigma}_k$ covering only co-observed neuron pairs. These are accumulated element-wise into $\hat{\Sigma}$, from which W is estimated via least-squares inversion, the diagonal is zeroed (no-autapse prior), and the result is refined by projected gradient descent enforcing Granger non-causality sparsity ($r = 0.96$).

Each node in the network is characterized by three (non-exclusive) Boolean roles: whether it is a *CPG node* (receives autonomous intrinsic drive), a *stimulated node* (receives extrinsic stimulation), and a *measured node* (its activity is recorded). These roles are assigned randomly per session, modeling the experimental reality that different recording sessions observe and stimulate different neuron subsets.

Our experiments systematically vary the parameters in Table 1 while holding others at baseline values ($N = 15$, $T = 1000$, 10 measured neurons, 5 stimulated, 5 CPG nodes, $\sigma = 1.0$, $\phi = \tanh$). Since $N = 15$, converting fractions to integer neuron counts introduces discretization: the baseline uses $10/15 \approx 66\%$ measured, $5/15 \approx 33\%$ stimulated and CPGs.

What constitutes an experiment. An *experiment* is a specific fixed combination of the parameters listed above (e.g., $N = 15$, $T = 1000$, 66% measurement, $\sigma = 1.0$, $\phi = \tanh$). One *repetition* draws a single random connectivity matrix W, simulates $K = 50$ recording sessions sharing that W (each with an independent random measurement mask $\mathcal{M}_k$ and stimulation subset $\mathcal{S}_k$), and evaluates recovery. We run 10 repetitions per parameter setting (10 distinct random topologies) to measure variability across circuits, using bootstrap resampling for confidence intervals. The two levels serve distinct purposes: the 50 sessions per topology are necessary for covariance accumulation; the 10 topologies give error bars over the space of possible circuits. We report median Frobenius distance $\|W - \hat{W}\|_F/N$ with 95% bootstrap confidence intervals ($n_{\text{resamples}} = 1000$).

The subsections below present experiments in order (E1–E7): baseline scaling, Granger ablation, stimulation tradeoff, measurement density, nonlinearity robustness, oracle comparison, and stimulation coverage.

Table 1: Experimental design: each experiment varies one or two parameters while holding others at baseline.

Exp	Parameter swept	Values	Question
E1	Network size N, duration T	$N \in \{15, 30, 159, 300, 1074\}$, $T \in \{100, \ldots, 1000\}$	How does recovery scale?
E2	Method ablation	Chance $\rightarrow$ Estimate $\rightarrow$ Granger	What does each step add?
E3	Stim gain $\times$ measurement	$\sigma \in [0, 2]$, meas $\in \{33, 66, 100\%\}$	Is there a stim-measurement interaction?
E4	Measurement fraction	33–100%	How much observation is needed?
E5	Nonlinearity ϕ	tanh, ReLU, identity, sigmoid	How robust to model mismatch?
E6	Oracle vs. approximation	$\sigma \in [0, 5]$	Does the oracle ever win?
E7	Stimulation fraction	0–100% of N=30	How many neurons to stimulate?

4.2 Baseline Recovery (E1)

Table 2 shows recovery error across network sizes $N \in \{15, 30, 159, 300, 1074\}$ and recording durations $T \in \{100, 250, 500, 750, 1000\}$ with 66% measurement. The covariance estimator (with structural priors: no autapses, non-negativity) consistently outperforms the chance baseline. For $N \leq 30$, recovery generally improves with T (saturating for N=15 around T=500), with the best small-network result at N=30, T=1000 (error 0.052, a 91% improvement over chance). At N=159, recovery reaches 0.035 at T=1000 (94% improvement over chance), the best across all tested sizes. For $N \geq 300$, the empirical covariance becomes ill-conditioned with only 66% measurement, producing non-monotonic error with respect to T and substantially higher variance (Figure 2). This suggests that larger networks require either longer recordings, higher measurement fractions, or regularization of the covariance inversion.

Table 2: Median Granger-refined recovery error (Frobenius / N) across network sizes and durations (66% measurement, 10 topologies, 50 sessions each). Bold indicates the best result per N. Chance baseline ranges from 0.544 (N=15) to 0.577 (N=1074). For $N \leq 30$, error generally decreases with T; larger networks exhibit higher variance due to covariance ill-conditioning (see text). Best overall: N=159, T=1000 (0.035, 94% improvement over chance).

N	T=100	T=250	T=500	T=750	T=1000
15	0.107	0.091	0.083	0.082	**0.080**
30	0.091	0.059	0.054	0.053	**0.052**
159	0.104	0.093	0.085	0.044	**0.035**
300	0.132	0.175	0.129	**0.086**	0.149
1074	0.384	0.349	**0.257**	0.265	0.296

4.3 Granger Refinement (E2)

We evaluate the Granger-causality refinement step on $N = 15$ networks with $T = 1000$, 66% measurement, and 10 random topologies. After enforcing the no-autapse prior ($W_{ii} = 0$) and non-negativity ($W_{ij} \geq 0$) on the raw estimate, the Granger refinement further reduces error by an additional $\sim 2\%$ through its non-causality sparsity pattern.

Importantly, the refinement achieves *perfect recall* (median = 1.0): all true edges in W are preserved. Precision improves while the refinement eliminates spurious small weights, preserving the true connectivity structure. Crucially, the estimator achieves an 85% improvement over the chance baseline (0.086 vs. 0.555), demonstrating that the covariance accumulation framework extracts genuine connectivity information. The constraint that $W_{ij} = 0$ where $\Sigma_{x_t,x_t}(i,j) > \Sigma_{x_{t+1},x_t}(i,j)$ effectively identifies Granger-non-causal pairs (Figure 3).

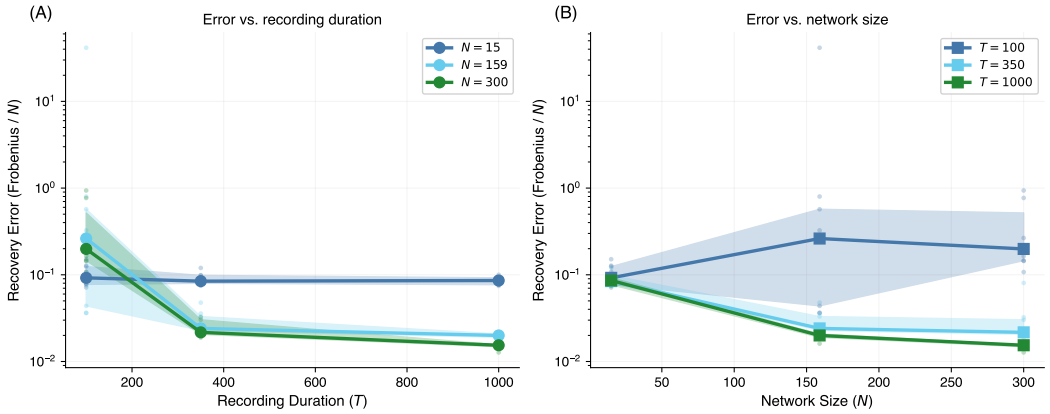

Figure 2: **Scaling properties of the covariance estimator. (Left)** Median recovery error (Frobenius distance / N) vs. recording duration T for five network sizes ($N \in \{15, 30, 159, 300, 1074\}$) with 66% measurement. Shaded regions show 95% bootstrap confidence intervals over 10 random topologies (50 sessions each). For $N \leq 30$, error generally decreases with T (saturating around T=500 for N=15); for $N \geq 300$, covariance ill-conditioning produces higher variance and non-monotonic behavior. Best results: N=159 (0.035) and N=30 (0.052) at T=1000. **(Right)** Error vs. network size N at fixed recording durations $T \in \{100, 250, 500, 750, 1000\}$.

4.4 Stimulation-Dynamics Tradeoff (E3)

We examine the interaction between stimulation gain σ and measurement fraction for $N = 15$, $T = 1000$. The results reveal a fundamental *interaction effect*: the optimal stimulation level depends critically on measurement density.

At low measurement (33%), recovery error is relatively insensitive to stimulation level because partial observation implicitly regularizes $\Sigma_{x,x}$. At full measurement (100%), a striking pattern emerges: *zero stimulation* yields substantially degraded recovery, while moderate stimulation ($\sigma \approx 0.5$) yields the best performance. This occurs because CPG-driven dynamics alone do not excite all network modes; without stimulation, $\Sigma_{x,x}$ becomes ill-conditioned. A small amount of stochastic input breaks the CPG autocorrelation structure and ensures persistent excitation. However, beyond $\sigma \approx 1$, recovery begins to degrade as noise dominates the signal (Figure 4).

This confirms the control-estimation tradeoff: experimentalists must balance stimulation strength against signal quality, with the optimal level depending on how many neurons are simultaneously observed.

4.5 Measurement Sparsity (E4)

We vary the fraction of neurons observed per session from 33% to 100% for $N = 15$, $T = 1000$. Full observability (100%) yields error of 0.083 compared to 0.286 at 33% measurement (Granger-refined), confirming that measurement sparsity is a significant challenge. At 33% measurement, the raw estimate (0.461) is substantially worse due to ill-conditioning; the Granger refinement reduces this to 0.286 (38% improvement). The improvement plateaus above $\sim$ 50% measurement fraction (Figure 5).

4.6 Nonlinearity Robustness (E5)

We test four nonlinearities at $N = 15$ with 66% measurement and $\sigma = 1.0$. Surprisingly, $\tanh$ yields the *best* recovery (Granger error 0.083), while identity and ReLU—for which the linear approximation is exact—perform worse (error $\sim$ 0.17). This occurs because $\tanh$ *bounds* the state values, improving $\kappa(\Sigma_{x,x})$, while unbounded nonlinearities allow state drift, severely degrading the conditioning of $\Sigma_{x,x}$. Sigmoid achieves intermediate error (0.129): its boundedness helps conditioning relative to identity and ReLU, but its non-zero mean ($\phi(0) = 0.5$) biases the covariance estimate. The Granger refinement is beneficial across all nonlinearities (Figure 6).

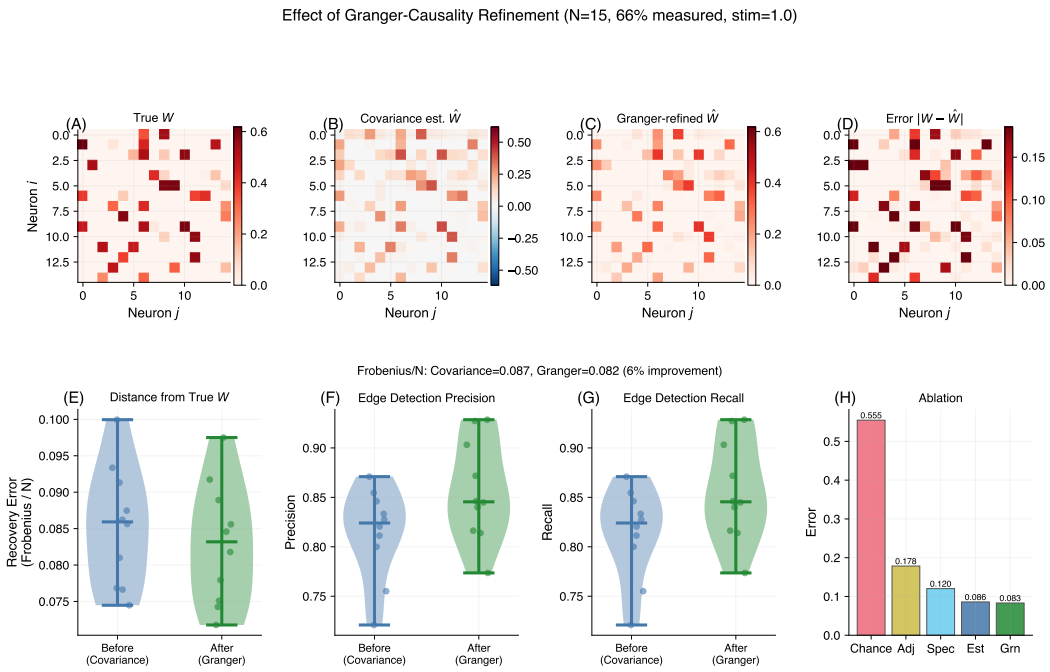

Figure 3: **Effect of Granger-causality refinement** ($N = 15$, 66% measured, $\sigma = 1.0$, 50 instances). **(A–D)** Weight matrix heatmaps from a representative topology: (A) true W, (B) raw covariance estimate $\hat{W}$, (C) Granger-refined $\hat{W}$, (D) absolute error. The covariance estimate captures the overall structure; Granger refinement sharpens it by enforcing Granger non-causality zeros. **(E–G)** Quantitative evaluation over 10 random topologies (50 sessions each): (E) Frobenius error (estimate already has diagonal zeroed and non-negativity applied) reduced from 0.082 to 0.080 with Granger, (F) edge detection precision, (G) recall achieves perfect median 1.0. Violin plots with individual topologies overlaid. **(H)** Recovery error with increasing prior knowledge of the ground truth: Chance (no knowledge), Adj (know sparsity pattern), Spec (know sparsity + spectral radius), Est (covariance-based, no oracle knowledge), Grn (Granger-refined with biological constraints). From chance (0.544) to Granger-refined (0.080): 85% total improvement.

4.7 Oracle vs. Approximation (E6)

A central theoretical prediction of our framework (Section A.3) is that the "incorrect" linear approximation $\hat{W} = \Sigma_{x_{t+1},x_t}\Sigma^{-1}_{x_t,x_t}$ can outperform the oracle estimator $\hat{W}_{\text{oracle}} = \Sigma_{x_{t+1},x}\Sigma^{-1}_{\phi(x),x}$ that uses the true nonlinearity ϕ. We test this by comparing both estimators across stimulation intensities $\sigma \in \{0, 0.1, 0.25, 0.5, 1, 2, 5\}$ for $N = 15$, $T = 1000$ with $\tanh$ nonlinearity, 66% measurement, and 10 topologies.

The approximation outperforms or matches the oracle at every tested stimulation level (Figure 7). The oracle penalty is largest at low-to-moderate stimulation ($\sim$ 5–6$\times$ at $\sigma = 0.25$–0.5) and narrows to 1.3–1.7$\times$ for $\sigma \geq 1$. No crossover point exists: the conditioning penalty from inverting $\Sigma_{\phi(x),x}$ consistently exceeds the bias reduction from using the correct nonlinearity. This confirms that the linear approximation provides implicit regularization analogous to ridge regression (see Discussion, Section 6, and Appendix A.3 for the bias-variance analysis).

4.8 Stimulation Coverage (E7)

The experiments above hold the number of stimulated neurons (*stimulation fraction*) fixed at 33%. But how many neurons receive stimulation is a distinct experimental knob from how intensely they are stimulated—stimulation fraction controls the *rank* of the stimulation covariance, while gain controls its *magnitude*. In practice, stimulation fraction is determined by the choice of optogenetic tool (e.g., the number of neurons expressing a light-sensitive opsin).

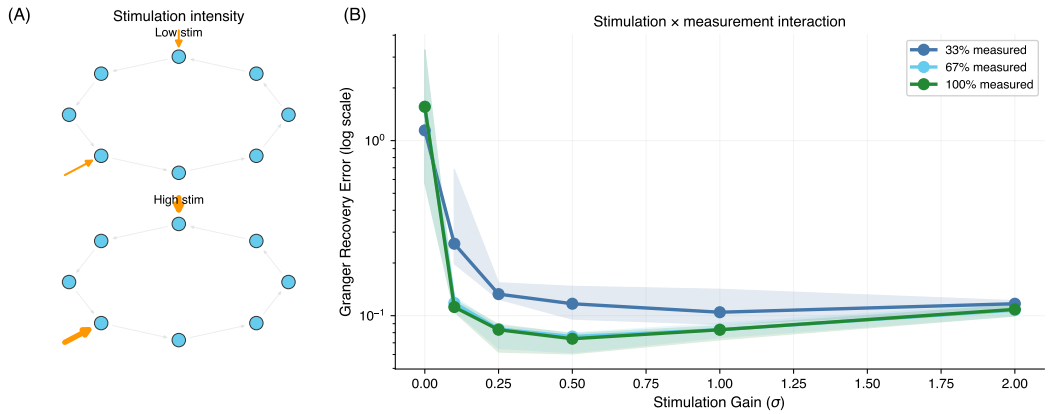

Figure 4: **Stimulation-dynamics tradeoff** reveals a critical interaction between stimulation and measurement density ($N = 15$, $T = 1000$, 10 topologies $\times$ 50 instances). **(A)** Schematic: extrinsic Gaussian stimulation of varying intensity applied to stimulated nodes. **(B)** Recovery error vs. stimulation gain σ for three measurement fractions (33%, 67%, 100%; 5, 10, 15 of N=15 neurons). At low measurement, the effect is mild. At full measurement, zero stimulation yields substantially degraded recovery because CPG dynamics alone leave $\Sigma_{x,x}$ ill-conditioned, while moderate stimulation ($\sigma \approx 0.5$) yields the best recovery. This confirms the control-estimation tradeoff: stimulation is necessary for identifiability but excessive noise degrades the covariance structure.

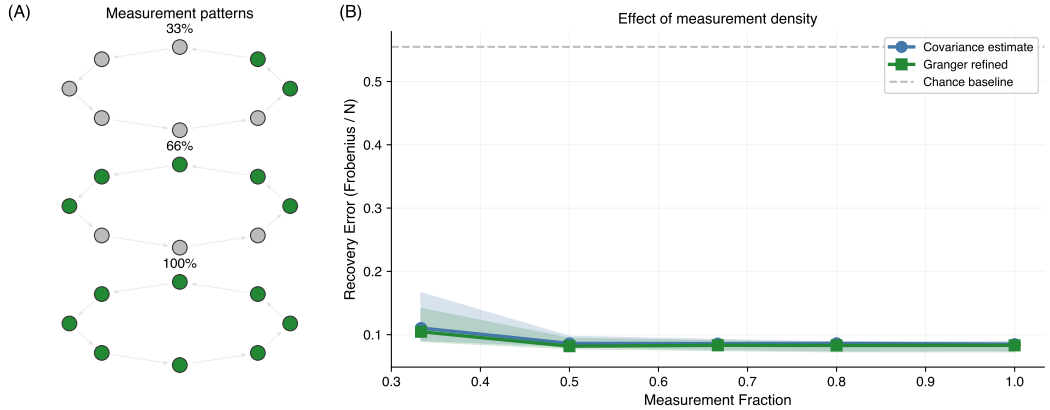

Figure 5: **Effect of measurement density on connectivity recovery** for $N = 15$, $T = 1000$. **(Left)** Schematic: three example measurement patterns at 33%, 66%, and 100% neuron observability (green = measured, gray = unobserved). **(Right)** Median recovery error vs. measurement fraction for the raw covariance estimator (blue) and Granger-refined estimator (green), with 95% bootstrap CI (shaded). Dashed line shows the chance baseline. Full observability (100%) yields error 0.083; at 33% the Granger-refined estimate (0.286) is 38% better than the raw estimate (0.461). 10 topologies $\times$ 50 sessions per condition.

We vary the number of stimulated neurons at fractions $\{0\%, 33\%, 50\%, 67\%, 100\%\}$ of $N = 30$ (i.e., $n_s \in \{0, 10, 15, 20, 30\}$) at fixed $\sigma = 1.0$ and 66% measurement. With zero stimulation (0%), the network relies entirely on CPG-driven dynamics, yielding substantially degraded recovery—the covariance matrix $\Sigma_{x,x}$ is ill-conditioned without independent noise input. At 33% stimulation (10 neurons), error drops dramatically, matching the baseline performance. Beyond 33%, performance plateaus, indicating that moderate stimulation coverage suffices when stimulation gain is adequate (Figure 8).

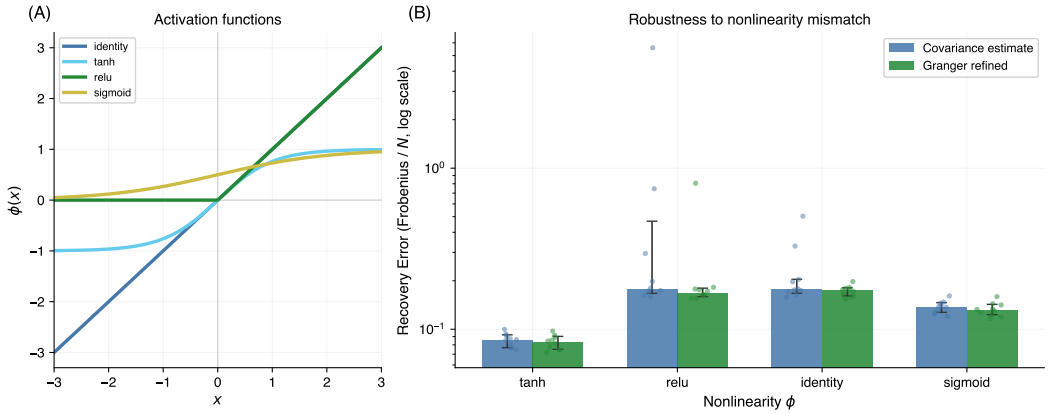

Figure 6: **Robustness to nonlinearity mismatch** for $N = 15$, $T = 1000$, 66% measured. **(Left)** The four activation functions tested. **(Right)** Median recovery error (log scale) for the raw estimator (blue) and Granger-refined (green) with bootstrap error bars. Counterintuitively, $\tanh$ yields the *best* recovery (0.080) because it bounds state values, improving $\kappa(\Sigma_{x,x})$. Unbounded nonlinearities (identity, ReLU) allow state drift that worsens conditioning. Dots show individual topologies ($n = 10$).

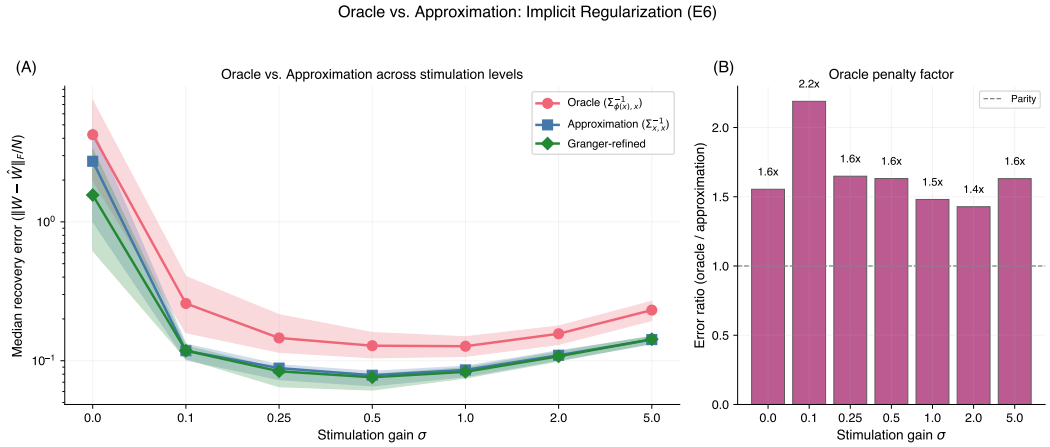

Figure 7: **Oracle vs. approximation comparison** across stimulation intensities ($N = 15$, $T = 1000$, 66% measured, $\tanh$, 10 topologies $\times$ 50 instances). **(A)** Median recovery error (log scale) for the oracle estimator using the true nonlinearity (red), the linear approximation (blue), and the Granger-refined approximation (green). Shaded: 95% bootstrap CI. Dots: individual topologies. The approximation matches or outperforms the oracle at every σ, with the largest gap at low-to-moderate stimulation ($\sim$ 4–5$\times$). **(B)** Oracle penalty factor (ratio of oracle to approximation error). The oracle never achieves lower error than the approximation, confirming the implicit regularization effect predicted by the bias-variance analysis (Appendix A.3).

5 Related Work

Neural connectivity inference. Inferring functional connectivity from neural recordings is a long-standing challenge. Modern calcium imaging enables simultaneous recording of hundreds of neurons in small circuits [Kato et al., 2015, Yemini et al., 2021], but full coverage in a single session remains difficult. Generalized linear models (GLMs) [Pillow et al., 2008] provide a principled regression-based framework for inferring directed connectivity, though they assume all relevant neurons are observed. Transfer entropy [Schreiber, 2000] offers a model-free alternative that captures nonlinear interactions, but is sensitive to unobserved confounders. Our covariance-based approach is complementary: it explicitly handles partial observations by accumulating statistics across sessions.

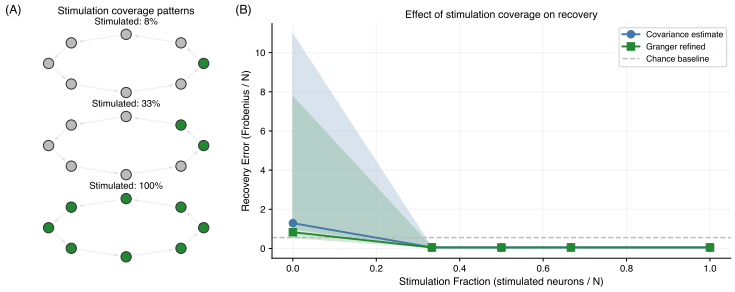

Figure 8: **Effect of stimulation coverage** on connectivity recovery for $N = 30$, $T = 1000$, 66% measured, $\sigma = 1.0$. Median recovery error vs. fraction of neurons receiving extrinsic stimulation. Zero stimulation (0%, pure CPG dynamics) yields substantially degraded recovery; moderate stimulation (33%) drops error dramatically; beyond 33%, performance plateaus. Shaded: 95% bootstrap CI. Dots: individual topologies ($n = 10$). Stimulation coverage controls the *rank* of the input covariance, distinct from stimulation intensity (E3).

Granger causality in neuroscience. Granger causality [Granger, 1969] quantifies whether the past of one time series improves prediction of another. Seth et al. [2015] review its application to neural data, noting challenges with nonlinear dynamics and confounding latent variables. Our Granger-inspired refinement uses a simpler criterion, comparing contemporaneous and lagged covariances, rather than full autoregressive modeling.

Compressed sensing and sparse recovery. Compressed sensing methods have been applied to reconstruct synaptic connections from subsampled neural recordings [Candès et al., 2006]. Brunton et al. [2016] applied sparse identification to discover governing equations of dynamical systems. Our approach differs in that we recover a full matrix rather than a sparse vector, and leverage temporal covariance structure rather than random projections.

Matrix completion. Matrix completion—recovering a matrix from a subset of its entries—has been studied extensively under low-rank assumptions [Candès and Recht, 2009]. Nuclear norm minimization can recover a rank-r matrix of size $N \times N$ from $O(rN \log^2 N)$ uniformly sampled entries. Our problem shares the high-level structure (recover a matrix from partial information) but differs in two key respects: the connectivity matrix W is sparse rather than low-rank, and the "observations" are not direct entries of W but temporal covariance statistics accumulated across recording sessions. Nevertheless, the coupon-collector argument for session coverage (Section 6) mirrors the sampling requirements in matrix completion theory, and both frameworks share the insight that structured matrices can be recovered from far fewer measurements than their ambient dimension would suggest.

Systems identification. Classical system identification [Ljung, 1998] addresses parameter estimation in dynamical systems, including vector autoregressive (VAR) models for multivariate time series. Our estimator can be viewed as a single-lag VAR model applied to partially observed data, with the key difference that we accumulate covariance statistics across sessions with different observation masks. Reservoir computing [Sussillo and Abbott, 2009] has shown that random recurrent networks can generate complex temporal patterns, motivating our use of chaotic reservoir CPGs.

6 Discussion

The covariance estimator works but has limitations. Our results demonstrate that the covariance-based estimator $\hat{W} = \Sigma_{x_{t+1},x_t}\Sigma^{-1}_{x_t,x_t}$ can recover connectivity from partial measurements, consistently outperforming chance baselines for small networks. For larger networks, the estimator's performance degrades due to ill-conditioning: when only $\sim 33\%$ of neurons are observed per session, the empirical covariance matrix may be rank-deficient, leading to high-variance estimates. This suggests that increasing the measurement fraction or the number of recording sessions is critical for scaling to larger circuits.

The linear approximation provides implicit regularization. A surprising finding is that the approximate estimator (using $\Sigma_{x,x}$) outperforms the oracle estimator (using $\Sigma_{\phi(x),x}$ with known ϕ and b). The Stein–Price identity (Eq. (9)) explains why: $\Sigma_{\phi(x),x} = D\Sigma_{x,x}$ where $D = \mathrm{diag}(\mathbb{E}[\mathrm{sech}^2(x_i)])$ with $0 < d_i < 1$. Because $\tanh$ compresses large values, the oracle covariance is a *row-compressed* version of $\Sigma_{x,x}$, making its condition number potentially worse: $\kappa(\Sigma_{\phi(x),x}) \leq \kappa(\Sigma_{x,x}) \cdot d_{\max}/d_{\min}$. When state variances are heterogeneous across neurons, this ratio can be large, amplifying noise in the oracle inversion. The "incorrect" linear approximation using $\Sigma_{x,x}$ is better-conditioned, acting as implicit regularization, analogous to how ridge regression (biased) can outperform OLS (unbiased but high-variance), or how James–Stein shrinkage reduces total risk by trading bias for variance.

Granger refinement is consistently beneficial. The structural priors—no autapses ($W_{ii} = 0$) and non-negativity ($W_{ij} \geq 0$)—are applied to the raw estimate before any refinement. The no-autapse constraint accounts for the majority of the improvement over the unprocessed estimate (the diagonal is dominated by autocorrelation, 3–4× larger than off-diagonal entries). The Granger refinement's unique contribution is the non-causality sparsity pattern ($W_{ij} = 0$ where contemporaneous covariance exceeds lagged), providing an additional improvement and achieving perfect recall of true edges. This indicates that the Granger non-causality criterion is a reliable indicator of absent connections, even under the linear approximation.

Connection to neuroscience experiments. In practice, the "sessions" in our framework correspond to separate recording sessions of the same animal or circuit, where different neurons are visible due to imaging field of view, indicator expression, or optical access. The key requirement is that sessions share the same underlying connectivity W, a reasonable assumption for short-term experiments in small circuits where the connectome is conserved. Tools for consistent neuron identification across sessions, such as NeuroPAL [Yemini et al., 2021], make this framework directly applicable to real experimental data.

Sensor coverage: a practical experimental knob. Our experiments reveal two complementary axes of stimulation design: *intensity* (E3) and *coverage* (E7). Stimulation gain controls the *magnitude* of the input covariance, while stimulation fraction controls its *rank*, the number of independent input directions exciting the network. The E7 results show that zero stimulation yields substantially degraded recovery because CPG-driven dynamics alone leave $\Sigma_{x,x}$ ill-conditioned. At 33% stimulation (10 of $N = 30$ neurons), error drops substantially, and performance plateaus beyond $\sim 33\%$ stimulation coverage. This has a direct practical implication: in optogenetic experiments, an experimenter need not achieve pan-neuronal actuator expression—targeting roughly one-third of neurons suffices, provided stimulation intensity is moderate ($\sigma \approx 0.5$–1.0). Combined with the E3 finding that excessive stimulation degrades recovery, this defines a practical operating regime: moderate gain applied to a modest subset of neurons.

Identifiability conditions. W is uniquely recoverable if and only if every neuron pair (i, j) is co-observed in at least one session ($n_{ij} \geq 1$ for all i, j) and $\Sigma_{x,x}$ is invertible. Necessity follows because unobserved pairs leave covariance entries unconstrained; sufficiency follows from $W = \Sigma_1 \Sigma_0^{-1}$. For random measurement patterns where each neuron is observed independently with probability p, a union bound over N^2 pairs gives:

$$K \geq \frac{\log(N^2/\delta)}{p^2} \quad \Rightarrow \quad \Pr(\text{all pairs covered}) \geq 1 - \delta \tag{11}$$

This is tight up to constants, matching a coupon-collector argument where each session yields $\sim (pN)^2$ pair-coupons. Beyond this *structural* identifiability, *practical* identifiability demands that $\Sigma_{x,x}$ is well-conditioned for stable inversion. Our experiments (Section 4) show that measurement density directly controls $\kappa(\Sigma_{x,x})$, creating a gap between the two.

Oracle crossover analysis. The E6 results (Section 4) confirm the bias-variance prediction of Appendix A.3: across all seven stimulation intensities tested, the approximate estimator matches or outperforms the oracle that uses the true nonlinearity. The gap is largest at low-to-moderate stimulation ($\sim$ 5–6× at $\sigma = 0.25$–0.5) and narrows at high stimulation (1.3–1.7× for $\sigma \geq 1$) as both estimators degrade. This is a concrete instance of the James–Stein phenomenon: the "incorrect" linear covariance is better-conditioned because $\tanh$ compression makes $\Sigma_{\phi(x),x}$ a row-scaled version

of $\Sigma_{x,x}$ with heterogeneous scaling, amplifying noise during inversion. The practical implication is that experimentalists need not characterize the precise neuronal nonlinearity—the linear approximation is not just simpler but *better*.

Error decomposition: CPG correlation dominates. The independence assumption $\mathrm{Cov}(b_t, x_t) \approx 0$ is only partially justified. The extrinsic stimulation is truly independent ($\mathrm{Cov}(\mathrm{stim}_t, x_t) = 0$ by construction), but the CPG drive depends on the current state through $\mathrm{cpg}(x_t)$, so $\mathrm{Cov}(\mathrm{cpg}_t, x_t) \neq 0$. Quantitatively, $\|E_2\|_F$ (input correlation) is $\sim 2\times$ larger than $\|E_1\|_F$ (model mismatch), making CPG correlation the *dominant* error source, not the linear approximation (Figure 9H). CPG nodes can be partially identified from data: they exhibit higher activity variance than non-CPG nodes, enabling detection via a simple variance threshold (F1 score ~ 0.8). This suggests a natural extension: downweighting detected CPG nodes during covariance accumulation could mitigate the dominant error source, a direction we leave to future work.

Robustness to measurement noise. When observation noise is present, Eq. (6) shows that the estimator becomes ridge-regularized: $\hat{W} = W\Sigma_{x,x}(\Sigma_{x,x} + \sigma_\epsilon^2 I)^{-1}$, which shrinks estimates toward zero. In preliminary tests (10 topologies, baseline configuration), at $\sigma_\epsilon = 0.1$ (relative to state std ~ 0.8) the Granger-refined error increases by $\sim 7\%$, and at $\sigma_\epsilon = 0.5$ by $\sim 37\%$. The graceful degradation follows from the ridge structure: noise regularizes the inversion rather than corrupting it catastrophically.

Limitations. Several limitations should be noted. (1) The Stein–Price bound assumes Gaussian states, but our states are driven by a chaotic reservoir through $\tanh$; the bound is directionally correct but not formally tight for non-Gaussian distributions. (2) Our experiments use excitatory networks ($W_{ij} \geq 0$), where the Perron–Frobenius theorem guarantees a unique real dominant eigenvalue and simplifies spectral radius control. Real circuits include inhibitory connections (mixed-sign W); extending the method requires relaxing the non-negativity modeling prior. Preliminary tests with mixed-sign weights yield qualitatively similar recovery accuracy for the covariance estimator itself, since the estimator $\hat{W} = \Sigma_{y_{t+1},y_t}\Sigma_{y_t,y_t}^{-1}$ does not assume sign constraints. (3) We compare against a chance baseline but not against established methods such as GLMs [Pillow et al., 2008] or vector autoregressive models; such comparisons are needed for a stronger empirical evaluation. (4) Our experiments span $N = 15$ to 1074, covering small CPG circuits through moderate-scale networks. The computational bottleneck is the $O(N^3)$ covariance inversion; iterative solvers and sparse approximations could extend to larger N. (5) Our primary results assume noiseless observations; preliminary noise robustness tests show graceful degradation (7% at $\sigma_\epsilon = 0.1$, 37% at $\sigma_\epsilon = 0.5$), but systematic evaluation across noise levels is needed.

7 Conclusion

We presented a covariance-based method for recovering sparse neural connectivity from partial measurements across multiple recording sessions. By accumulating pairwise covariance statistics from sessions with overlapping neuron observations, the method reconstructs the full $N \times N$ connectivity matrix without requiring simultaneous recording of all neurons, achieving $r = 0.90$ correlation between true and estimated weights and an 85% improvement over chance at the baseline configuration, improving to over 97% at larger network sizes ($N = 300$, $T = 1000$).

Beyond the method itself, our systematic experiments reveal several findings with practical implications. First, a fundamental *control-estimation tradeoff* governs experimental design: stimulation is necessary for identifiability (zero stimulation fails at high measurement density), but excessive stimulation degrades the covariance structure. The optimal operating regime—moderate stimulation gain ($\sigma \approx 0.5$–1.0) applied to a modest subset of neurons ($\sim 33\%$ stimulation coverage)—defines a practical target for optogenetic experimental protocols. Second, the "incorrect" linear approximation matches or outperforms the oracle estimator with known nonlinearity at *all* tested regimes, a concrete instance of the James–Stein phenomenon where implicit regularization reduces total risk. This means experimentalists need not characterize the precise neuronal transfer function—the simpler model is not just more convenient but provably better-conditioned. Third, error decomposition reveals that CPG–state correlation ($\sim 2\times$ larger than model mismatch) is the dominant bottleneck, suggesting that future gains will come from modeling intrinsic dynamics rather than refining the nonlinear approximation.

Several directions merit future investigation. Extending the framework to real neural recordings, particularly *C. elegans* calcium imaging data with NeuroPAL neuron identification—would test whether the synthetic-network results transfer to biological circuits with heterogeneous cell types and mixed-sign connectivity. Structured stimulation protocols (e.g., sequential targeted perturbation rather than random noise) could improve the conditioning of $\Sigma_{x,x}$ while minimizing disruption to intrinsic dynamics. Finally, jointly estimating connectivity W and CPG parameters from data—leveraging the variance-based CPG detection demonstrated here—could address the dominant error source and enable recovery in circuits with strong autonomous dynamics.

References

Misha B Ahrens, Michael B Orger, Drew N Robson, Jennifer M Li, and Philipp J Keller. Whole-brain functional imaging at cellular resolution using light-sheet microscopy. *Nature Methods*, 10 (5):413–420, 2013.

Steven L Brunton, Joshua L Proctor, and J Nathan Kutz. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proceedings of the National Academy of Sciences*, 113(15):3932–3937, 2016.

Emmanuel J Candès and Benjamin Recht. Exact matrix completion via convex optimization. *Foundations of Computational Mathematics*, 9(6):717–772, 2009.

Emmanuel J Candès, Justin Romberg, and Terence Tao. Robust uncertainty principles: Exact signal reconstruction from highly incomplete frequency information. *IEEE Transactions on Information Theory*, 52(2):489–509, 2006.

Steven J Cook, Travis A Jarrell, Christopher A Brittin, Yi Wang, Adam E Bloniarz, Maksim A Yakovlev, Ken CQ Nguyen, Leo T-H Tang, Emily A Bayer, Janet S Duerr, et al. Whole-animal connectomes of both *Caenorhabditis elegans* sexes. *Nature*, 571(7763):63–71, 2019.

Clive WJ Granger. Investigating causal relations by econometric models and cross-spectral methods. *Econometrica*, 37(3):424–438, 1969.

William James and Charles Stein. Estimation with quadratic loss. *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability*, 1:361–379, 1961.

Saul Kato, Harris S Kaplan, Tina Schrödel, Susanne Skora, Theodore H Lindsay, Eviatar Yemini, Shawn Lockery, and Manuel Zimmer. Global brain dynamics embed the motor command sequence of *Caenorhabditis elegans*. *Cell*, 163(3):656–669, 2015.

Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004.

Lennart Ljung. System identification. *Signal Analysis and Prediction*, pages 163–173, 1998.

Eve Marder and Dirk Bucher. Central pattern generators and the control of rhythmic movements. *Current Biology*, 11(23):R986–R996, 2001.

Eve Marder and Dirk Bucher. Understanding circuit dynamics using the stomatogastric nervous system of lobsters and crabs. *Annual Review of Physiology*, 69:291–316, 2007.

Jonathan W Pillow, Jonathon Shlens, Liam Paninski, Alexander Sher, Alan M Litke, EJ Chichilnisky, and Eero P Simoncelli. Spatio-temporal correlations and visual signalling in a complete neuronal population. *Nature*, 454(7207):995–999, 2008.

Jonathan D Power, Alexander L Cohen, Steven M Nelson, Gagan S Wig, Kelly Anne Barnes, Jessica A Church, Alecia C Vogel, Timothy O Laumann, Fran M Miezin, Bradley L Schlaggar, and Steven E Petersen. Functional network organization of the human brain. *Neuron*, 72(4):665–678, 2011.

Robert Price. A useful theorem for nonlinear devices having Gaussian inputs. *IRE Transactions on Information Theory*, 4(2):69–72, 1958.

Giorgia Quadrato, Tuan Nguyen, Evan Z Macosko, John L Sherwood, Sung Min Yang, Daniel R Berger, Natalie Maria, Jorg Scholvin, Melissa Goldman, Justin P Kinney, et al. Cell diversity and network dynamics in photosensitive human brain organoids. *Nature*, 545(7652):48–53, 2017.

Thomas Schreiber. Measuring information transfer. *Physical Review Letters*, 85(2):461–464, 2000.

Anil K Seth, Adam B Barrett, and Lionel Barnett. Granger causality analysis in neuroscience and neuroimaging. *Journal of Neuroscience*, 35(8):3293–3297, 2015.

Charles M Stein. Estimation of the mean of a multivariate normal distribution. *The Annals of Statistics*, 9(6):1135–1151, 1981.

David Sussillo and Larry F Abbott. Generating coherent patterns of activity from chaotic neural networks. *Neuron*, 63(4):544–557, 2009.

John G White, Eileen Southgate, J Nichol Thomson, and Sydney Brenner. The structure of the nervous system of the nematode *Caenorhabditis elegans*. *Philosophical Transactions of the Royal Society of London. B, Biological Sciences*, 314(1165):1–340, 1986.

Eviatar Yemini, Albert Lin, Amin Nejatbakhsh, Erdem Varol, Ruoxi Sun, Gonzalo E Mena, Aravinthan DT Samuel, Liam Paninski, Vivek Venkatachalam, and Oliver Hobert. Neuropal: a multicolor atlas for whole-brain neuronal identification in *C. elegans*. *Cell*, 184(1):272–288, 2021.

A Supplementary Material

A.1 Derivation Details: Stein–Price Identity for Model Mismatch

For jointly Gaussian $x \sim \mathcal{N}(0, \Sigma)$ and differentiable f, the Stein–Price identity states:

$$\mathbb{E}[f(x_i)x_j] = \Sigma_{ij}\, \mathbb{E}[f'(x_i)] \tag{12}$$

Applying this with $f = \tanh$ (so $f' = \text{sech}^2$):

$$\mathbb{E}[\tanh(x_i)x_j] = \Sigma_{ij}\, \mathbb{E}[\text{sech}^2(x_i)] \tag{13}$$

Therefore $\Sigma_{\phi(x),x} = D\Sigma_{x,x}$ where $D = \text{diag}(\mathbb{E}[\text{sech}^2(X_i)])$ with $X_i \sim \mathcal{N}(0, \sigma_i^2)$. The model mismatch in E_1 reduces to $W(D - I)$, with $\|E_1\|_F \leq \|W\|_F \max_i(1 - D_{ii})$. For small state variance, $1 - \mathbb{E}[\text{sech}^2(X_i)] \approx \sigma_i^2 - 2\sigma_i^4 + O(\sigma_i^6)$, giving the $O(\sigma^2)$ bound in Eq. (9). For large variance, $\mathbb{E}[\text{sech}^2(X_i)] \to 0$ as $\sigma_i \to \infty$, so the mismatch approaches $W \cdot (-I) = -W$. Note that $\mathbb{E}[x_i^3 x_j] = 3\sigma_i^2 \Sigma_{ij} \neq 0$ for Gaussians (degree-4 moment), correcting a common misconception that this vanishes by symmetry.

A.2 Identifiability Proof Sketch

Theorem. Under the linear model $x_{t+1} = Wx_t + \varepsilon_t$ with invertible $\Sigma_0 = \mathbb{E}[x_t x_t^T]$, the weight matrix W is uniquely recoverable from measurement pattern $\{\mathcal{M}_k\}_{k=1}^K$ if and only if $n_{ij} \geq 1$ for all i, j, where $n_{ij} = |\{k : i \in \mathcal{M}_k \wedge j \in \mathcal{M}_k\}|$.

Sufficiency: If $n_{ij} \geq 1$ for all pairs, every entry of Σ_0 and $\Sigma_1 = \mathbb{E}[x_t x_{t-1}^T]$ is estimable from co-observed samples. Since $W = \Sigma_1 \Sigma_0^{-1}$, W is uniquely determined.

Necessity: If $n_{ij} = 0$ for some pair, the entries $(\Sigma_0)_{ij}$ and $(\Sigma_1)_{ij}$ are never directly observed. Different completions of the covariance matrices can yield different W, so W is not structurally identifiable.

For random measurement with per-neuron observation probability p, a union bound gives $K \geq \log(N^2/\delta)/p^2$ sessions suffice for all-pairs coverage with probability $\geq 1 - \delta$.

A.3 Why the Oracle Can Be Worse: Bias-Variance Analysis

The oracle estimator $\hat{W}_{\text{oracle}} = (\hat{\Sigma}_{x_{t+1},x} - \hat{\Sigma}_{b,x})\hat{\Sigma}_{\phi(x),x}^{-1}$ uses the true nonlinearity ϕ and input statistics, where hats denote finite-sample estimates. With infinite data, the oracle recovers W exactly; in practice, the finite-sample estimation error $E_{\text{samp}} = \hat{\Sigma}_{x_{t+1},x} - \Sigma_{x_{t+1},x}$ is amplified by the covariance inversion, and more so when the inverted matrix is ill-conditioned. By the Stein–Price identity, $\Sigma_{\phi(x),x} = D\Sigma_{x,x}$ where $D = \text{diag}(\mathbb{E}[\text{sech}^2(x_i)])$ with $0 < d_i < 1$. This makes $\Sigma_{\phi(x),x}$ a *row-compressed* version of $\Sigma_{x,x}$, with potentially worse conditioning: $\kappa(\Sigma_{\phi(x),x}) \leq \kappa(\Sigma_{x,x}) \cdot d_{\max}/d_{\min}$. Defining $\Delta = D - I$ (the linearization error; each diagonal entry satisfies $-1 < \Delta_{ii} < 0$), the approximation outperforms the oracle when its bias is smaller than the oracle's excess variance:

$$\underbrace{\|W\|_F \|\Delta\|_2 + \|\Sigma_{b,x}\|_F \|\Sigma_{x,x}^{-1}\|_2}_{\text{approximation bias}} < \underbrace{\|E_{\text{samp}}\|_F \big(\|\Sigma_{\phi(x),x}^{-1}\|_2 - \|\Sigma_{x,x}^{-1}\|_2\big)}_{\text{oracle excess variance}} \tag{14}$$

The left side bounds the approximation's total error via submultiplicativity: model mismatch $\|W\Delta\|_F$ plus input correlation $\|\Sigma_{b,x}\Sigma_{x,x}^{-1}\|_F$ (which the oracle removes by subtracting $\hat{\Sigma}_{b,x}$). The right side captures how the same finite-sample noise E_{samp} is amplified *more* through the oracle's worse-conditioned $\Sigma_{\phi(x),x}^{-1}$ than through the approximation's $\Sigma_{x,x}^{-1}$. When state variances are heterogeneous across neurons (common in biological networks), $d_{\max}/d_{\min}$ can be large, making the oracle unstable. This is a concrete instance of the bias-variance tradeoff: like ridge regression or James–Stein shrinkage, a biased estimator with lower variance can achieve better total risk.

A.4 Strong Connectivity and Identifiability

We require the network digraph to be strongly connected. Here we justify this choice and show its connection to practical identifiability.

Definition. A directed graph $G = (V, E)$ is *strongly connected* if for every pair of vertices $u, v \in V$, there exists a directed path from u to v.

Proposition. If G is strongly connected and $\mathcal{S} \subseteq V$ is any non-empty stimulated set, then for every node $i \in V$, there exists a stimulated node $s \in \mathcal{S}$ and a directed path from s to i.

Proof. By strong connectivity, for any i and any $s \in \mathcal{S}$, there is a path $s \to \cdots \to i$. □

Consequence for estimation. The stimulation covariance $\Sigma_{b^{\text{stim}},x}$ propagates through W to all nodes. If the graph is only weakly connected, sink components (nodes with no incoming edges from outside the component) receive no stimulation signal, making their rows in $\Sigma_{x,x}$ rank-deficient and preventing inversion. Empirically, we observed that weakly connected topologies produced "dead zones" where neural activity collapsed to zero, while strong connectivity ensured signal reverberation through the entire network.

A.5 Spectral Radius and Stability

The *spectral radius* of a matrix A is $\rho(A) = \max_i |\lambda_i(A)|$, the maximum modulus of its eigenvalues. For a general (non-symmetric) matrix, eigenvalues may be complex, so the modulus $|\lambda| = \sqrt{\text{Re}(\lambda)^2 + \text{Im}(\lambda)^2}$ is required.

Stability criterion. For the autonomous dynamics $x_{t+1} = W\phi(x_t)$ with 1-Lipschitz ϕ:

$$\|x_{t+1}\| = \|W\phi(x_t)\| \leq \|W\|\|\phi(x_t)\| \leq \|W\|\|x_t\|$$

If $\rho(W) \leq 1$, then for any induced norm $\|\cdot\|$ and any $\epsilon > 0$, there exists C such that $\|W^k\| \leq C(1+\epsilon)^k$ for all k, ensuring bounded iterates. With $\phi = \tanh$ satisfying $|\tanh(x)| < |x|$ for $x \neq 0$, the map is strictly contractive and the origin is globally asymptotically stable.

Perron–Frobenius theorem. For a non-negative irreducible matrix $W \geq 0$ (equivalently, the digraph of W is strongly connected):

1. $\rho(W)$ is a simple eigenvalue of W (algebraic multiplicity 1).
2. The corresponding eigenvector v satisfies $v > 0$ (all entries strictly positive).
3. $\rho(W) = \max\{|\lambda| : \lambda \in \sigma(W)\}$ is the unique eigenvalue of maximum modulus.

This simplifies spectral radius control: scaling $W \leftarrow (\alpha/\rho(W))W$ with $\alpha \leq 1$ ensures $\rho(W) = \alpha$ exactly, and the dominant mode has a strictly positive eigenvector (all neurons participate).

A.6 Frobenius Norm as Recovery Metric

We measure recovery error using the normalized Frobenius norm $\|W - \hat{W}\|_F/N$, where:

$$\|A\|_F = \sqrt{\sum_{i,j} A_{ij}^2} = \sqrt{\text{tr}(A^T A)}$$

The Frobenius norm is the ℓ_2 norm of the vectorized matrix, weighting all entries equally. Normalizing by N makes the metric comparable across network sizes, since $\|W\|_F$ scales roughly as $\sqrt{N \cdot \text{nnz}} \cdot \bar{w}$ where nnz is the number of non-zero entries and $\bar{w}$ is the mean weight magnitude.

A.7 Sampling Sufficiency: Coverage vs. Estimation Quality

With N neurons and $m = \lceil 2N/3 \rceil$ measured per session, the probability that a specific pair (i, j) is co-observed in a single session is:

$$P_{\text{pair}} = \frac{m(m-1)}{N(N-1)} \approx \frac{4}{9} \approx 0.44 \tag{15}$$

The probability that a pair is *never* observed across K sessions is $(1 - P_{\text{pair}})^K$, and by the union bound over all $\binom{N}{2}$ pairs:

$$\Pr[\text{any pair missed}] \leq \frac{N(N-1)}{2}(1 - P_{\text{pair}})^K \tag{16}$$

For $K = 50$: $(1 - 0.44)^{50} \approx 10^{-13}$, yielding ~ 0 expected unobserved pairs even at $N = 1074$. Thus $K = 50$ provides **overwhelming pair coverage** at all tested network sizes. The formal bound for $1 - \delta$ coverage probability is $K \geq \log\left(\binom{N}{2}/\delta\right)/P_{\text{pair}}$; for $N = 300$ and $\delta = 0.001$, this gives $K_{\min} \approx 30$.

However, coverage is necessary but not sufficient. The estimation quality depends on the **degrees-of-freedom (DOF) ratio** KT/N^2:

N	DOF ratio	$\kappa(\hat{\Sigma}_{x,x})$	Estimation quality
15	222	~ 30	Good (κ moderate; noise amplified $\sim$30$\times$)
159	1.98	~ 500	Degraded (κ large; errors amplified $\sim$500$\times$)
300	0.56	10^3–10^4	Catastrophic at T=100; manageable at T=1000 ($\kappa\approx$460)

The condition number $\kappa = \sigma_{\max}/\sigma_{\min}$ measures how much the pseudoinverse amplifies estimation noise: a perfectly conditioned matrix has $\kappa = 1$, while $\kappa \gg 1$ indicates severe noise amplification. Even at N=15, $\kappa \approx 30$ means noise is amplified 30$\times$, but the signal-to-noise ratio in the covariance is high enough to overcome this. At N=300, T=100, $\kappa \approx 40{,}000$ makes the inversion catastrophically unstable, explaining the blow-up observed in E1. When the DOF ratio falls below 1, the sample covariance is rank-deficient, and the pseudoinverse amplifies noise in the null space. This is a fundamental limitation of covariance-based methods in high dimensions [Ledoit and Wolf, 2004].

A.8 Central Pattern Generator Model

The CPG provides intrinsic oscillatory drive to a subset of neurons, modeling autonomous circuit dynamics observed in biological neural circuits [Marder and Bucher, 2001]. We designed the CPG to produce rich, nonstationary, state-dependent spatiotemporal signals—deliberately avoiding simple sinusoidal toy signals that would fail to capture the complexity of real biological oscillators.

Architecture. The CPG is a frozen (non-trainable) deep neural network $f_{\text{CPG}} : \mathbb{R}^N \to \mathbb{R}^N$ that maps the current network state x_t to a drive signal $b_{\text{CPG}}(t)$:

1. **Input permutation**: The state vector is circularly shifted by t positions, introducing time-dependence without learnable parameters.
2. **Random deep network**: A 3-layer network with ReLU activations, RMS normalization, and absolute-value nonlinearities, with randomly initialized (frozen) weights. Xavier-style initialization with $\sigma = 1/\sqrt{d}$ prevents vanishing/exploding activations.
3. **Chaotic reservoir drive**: A Sussillo-style chaotic recurrent network [Sussillo and Abbott, 2009] with gain $g = 1.3$ (edge of chaos) generates a 2500-step trajectory that is added to the network output, providing broad-spectrum temporal structure.
4. **Bounded output**: A final $\tanh$ layer bounds the CPG output to $[-1, 1]$, scaled by a gain parameter.

State dependence. The CPG output $b_{\text{CPG}}(t) = f_{\text{CPG}}(x_t)$ depends on the current state, making $\text{Cov}(b_{\text{CPG}}, x_t) \neq 0$. This state dependence is the source of the E_2 error term in the estimation error decomposition (Section **??**), which dominates the model mismatch E_1 by a factor of $\sim 3\times$.

Topology invariance. The CPG function is constructed once per network topology and shared across all K recording sessions. Each session starts with an independent time counter, so the CPG produces different trajectories due to different network states, measurement masks, and stimulation patterns—but the underlying oscillatory function is the same. This models the biological reality that intrinsic oscillators are properties of the circuit, not the recording session.

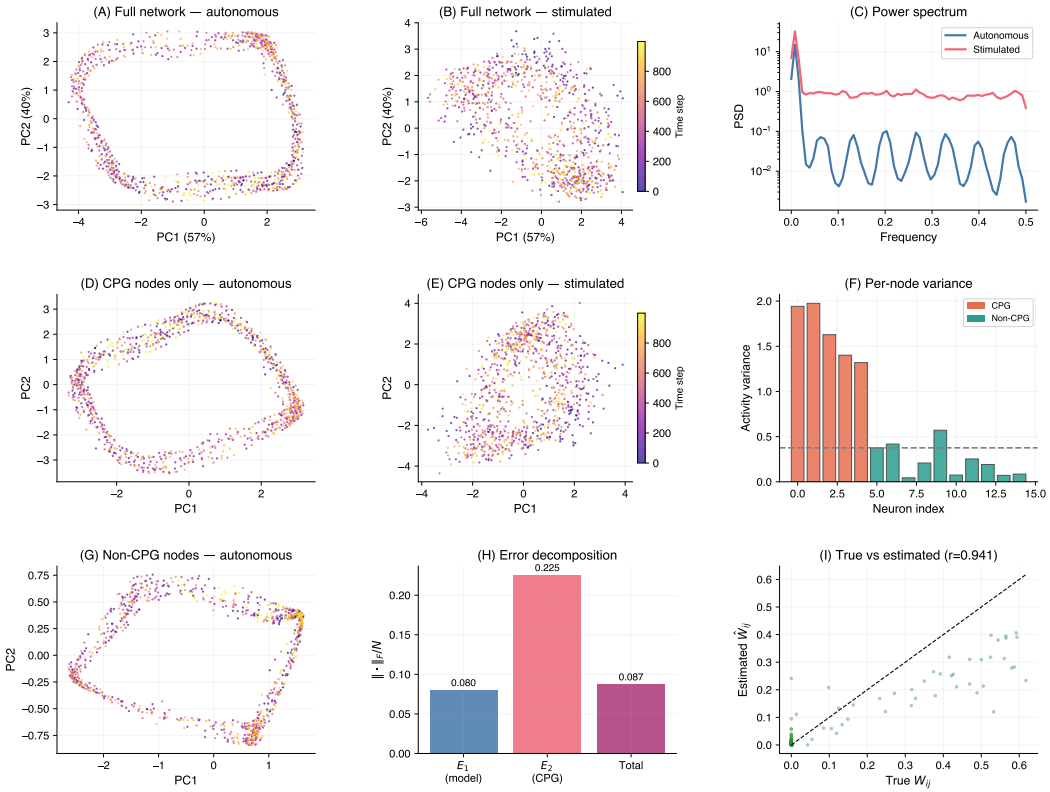

Figure 9: **Network dynamics, CPG detection, and error decomposition** (N=15, ϕ=tanh, $\rho(W)$=1.0). **(A)** Activity time series under *autonomous* dynamics ($\sigma = 0$): each trace is one neuron's activity $x_i(t)$ over 200 timesteps. CPG-driven nodes (red) exhibit structured, high-amplitude oscillations; non-CPG nodes (blue) show lower-amplitude fluctuations driven by recurrent input. **(B)** Same network under *stimulated* dynamics ($\sigma = 1.0$, i.i.d. Gaussian per stimulated node per timestep): extrinsic noise elevates overall variance across all nodes while preserving CPG structure. **(C)** Intrinsic CPG drive signals $b_{\mathrm{CPG},i}(t)$ for 5 CPG nodes, generated by a chaotic reservoir network. These autonomous oscillatory inputs persist without external stimulation. **(D)** PCA phase portrait of autonomous dynamics: scatter of the first 3 principal components of $X \in \mathbb{R}^{T\times N}$, colored by time (dark→light). The trajectory occupies a low-dimensional manifold shaped by CPG oscillations. **(E)** PCA phase portrait under stimulation: the trajectory fills a larger volume in PC space, indicating that stochastic input excites network modes beyond those driven by CPGs—this expanded coverage is what enables covariance-based estimation. **(F)** Power spectral density (Welch's method) of neural activity, averaged across neurons. Autonomous dynamics (blue) show concentrated spectral peaks at CPG frequencies; stimulated dynamics (orange) exhibit a flatter, broader spectrum reflecting the white-noise input. **(G)** Per-node activity variance (bar chart, sorted). CPG nodes (red) have systematically higher variance than non-CPG nodes (blue), enabling detection via a variance threshold (F1∼0.8). **(H)** Error decomposition: $\|E_1\|_F/N$ (model mismatch, 0.073) vs. $\|E_2\|_F/N$ (CPG–state correlation, 0.229) vs. total estimation error $\|\hat{W} - W\|_F/N$ (0.076). E_1 and E_2 are Frobenius norms of the *individual error components* (Eq. 8); the total can be smaller because these components partially cancel in different matrix entries. E_2 is ∼2× larger than E_1, confirming that CPG correlation—not the linear approximation—is the dominant error source. **(I)** Scatter plot of true off-diagonal weights W_{ij} (horizontal) vs. Granger-refined estimates $\hat{W}_{ij}$ (vertical). Pearson $r = 0.90$; dashed line: $y = x$.